



Two dimensional beam forming and steering based on silicon nitride 1×32 optical phased array

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Abstract: Optical phased arrays (OPAs) offer a promising solution for integrating beam focusing and steering functionalities on a chip. In this work, we present a 32-antenna OPA fully integrated on a commercially available silicon nitride (SiN) platform, operating in the wavelength range of 1500 nm to 1630 nm. We detail the design of the circuit and its components, emphasizing the development of a highly efficient phase shifter that requires 130 mW for a 2π phase shift. Testing the fabricated and assembled chip demonstrates a field of view of $24.28^\circ \times 9.4^\circ$ and a beam resolution of $0.66^\circ \times 0.082^\circ$. To the best of our knowledge, these values are the highest reported for a fully SiN-based OPA at 1550 nm.

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1. Introduction

A photonic integrated circuit (PIC) is a miniaturized photonic system composed of several elementary components, designed to serve a wide range of applications and functionalities [1]. PICs can be fabricated using various technological platforms, including silicon (Si), indium phosphide (InP), lithium niobate (LN), and silicon nitride (SiN), each offering unique advantages and disadvantages [1–4]. Si and SiN are particularly valued for their compatibility with CMOS fabrication technology, which significantly reduces production costs [2,3].

One potential application of PICs is the implementation of optical phased arrays (OPAs) for optical beam focusing and steering [5–9]. OPAs function on the same principle as phased array systems used in RADAR applications, but operate in the optical domain [10–12]. A phased array comprises an array of antennas emitting signals with tightly controlled phases. Through constructive interference, these signals can be focused into a single beam and steered [12,13]. Phase control of the emitted signals is achieved using dedicated phase shifters (PSs) before each antenna. Integrated OPAs offer several advantages over mechanical beam steering systems, including mechanical robustness, miniaturization, low power consumption, and cost-effectiveness [12–15].

OPAs are versatile and can be used in a variety of applications, such as light detection and ranging (LiDAR) and free-space optical communication [11,12,16]. For LiDAR application, OPAs are expected to provide a large field of view (FOV), high resolution, and low power consumption, all critical requirements for this application [11,12,17]. In LiDAR and data communication, where high output optical power is necessary for long-range transmission, the use of Si can be challenging due to its third-order non-linearity, which could specifically lead to two-photon absorption [5,6]. Conversely, the SiN platform offers unique properties like low non-linearity, a wide transparency spectrum (from visible to infrared), low propagation loss, smaller refractive index contrast ($n_{\text{SiN}} \sim 2$, $n_{\text{SiO}_2} \sim 1.5$), and a low thermo-optic coefficient [3,7,8,18]. These characteristics make SiN less susceptible to phase errors introduced during

fabrication, more stable against environmental temperature variations, and capable of supporting high optical power transmission [3,7].

Thanks to these advantages, SiN-based OPAs can incorporate a larger number of antennas, resulting in better beam resolution [3]. However, the low thermo-optic coefficient of SiN poses a disadvantage when integrating PSs, making them less efficient compared to their Si counterparts [9,19]. Additionally, significant heat dissipation and potential thermal crosstalk between PSs can complicate the system design [3].

To the best of our knowledge, this work presents the first OPA fully integrated on a SiN platform that can be controlled by dedicated PSs at 1550 nm. The state-of-the-art predominantly focuses on silicon-on-insulator (SOI) or hybrid SiN and SOI platforms [9,12,20]. In the latter case, although the platform performs satisfactorily, it requires complex fabrication processes involving multiple deposition and etching steps [21].

In this study, we report on the design, fabrication, and characterization of a 1×32 SiN OPA chip. The PSs integrated on this chip have been designed to obtain improved efficiency. The OPA achieves an FOV of $24.28^\circ \times 9.46^\circ$ with a beam full width at half maximum (FWHM) of $0.66^\circ \times 0.082^\circ$, improving over previously reported results for other fully SiN-based OPAs operating at 905 nm [8,22].

This paper is divided into three sections. The first section provides an overview of the chip's design, fabrication, and assembly, including the design of the PSs and OPA antennas. The second section discusses the beam steering principles, the characterization setup of the chip, and the obtained results. Finally, the conclusion section highlights the key findings of our study.

2. Design, fabrication and assembly

Various solutions have been proposed to split and redistribute the input signal to the multiple output antennas in an OPA. One approach is the cascaded method, where the input signal is continuously divided through a chain of directional couplers [13]. In this case, the coupling strength of each directional coupler must be precisely controlled to achieve the desired power distribution among the antennas. Each directional coupler is followed by a PS to control the output phase, with the effect of each PS adding to the previous ones. In theory, this allows all phase shifters to operate with a single driving signal (voltage or current). However, any discrepancies between the designed and fabricated circuits would require independent PS driving signals to compensate the phase error.

Another approach involves multiple light splitting, in which each antenna has a dedicated PS with a separate driving signal, resulting in a uniform power distribution among the output antennas. The advantage of this method over the cascaded approach lies in the light distribution to the antennas. In the cascaded method, achieving a uniform distribution requires careful design of each directional coupler's power drop. In contrast, the multiple splitting approach uses a simple cascade of 3-dB splitters for the distribution network. This work adopts the multiple splitting approach.

A simplified schematic of the proposed OPA circuit is shown in Fig. 1(a). As shown in the figure, an edge coupler is used to couple the light into the PIC and a polarization beam splitter (PBS) is employed to divert any TM polarized signal to a dedicated output coupler. Therefore, the TE signal is directed to the distribution network with 32 outputs, each connected to a PS and antenna. The output from each antenna is combined two at a time into an output edge coupler array. Using an edge coupler instead of a grating coupler prevents vertical light emission and its interference with the OPA. The combiner and output coupler array were introduced to potentially assist in the far-field calibration process detailed in Sec. 3.2.

The proposed OPA circuit was fabricated by LIGENTEC [23] in a multi-project wafer run using 800 nm thick SiN. The optical microscopy image of the OPA chip is demonstrated in Fig. 1(b). This technology comes with a comprehensive process design kit (PDK). The components from

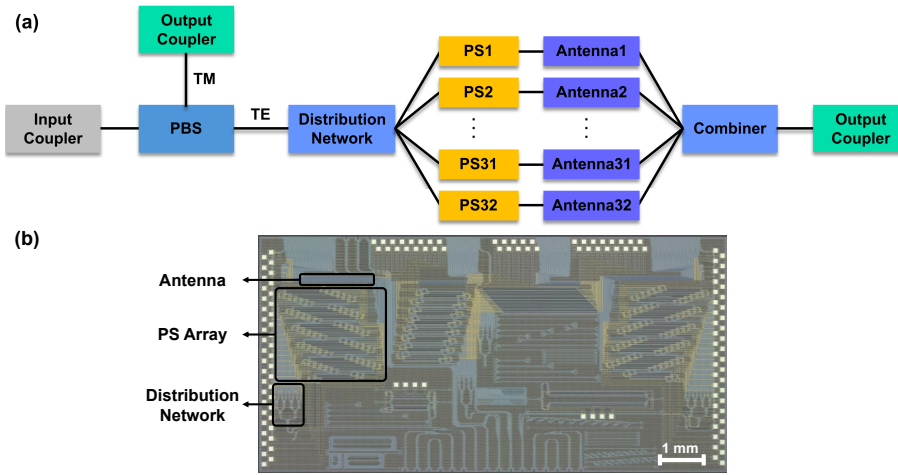


Fig. 1. (a) Simplified building block of the proposed OPA. (b) Optical microscopy image of the OPA chip.

the LIGENTEC PDK used in this work include edge couplers for effective light coupling from a single mode fiber (SMF), PBS, and multi-mode interference (MMI) 1×2 splitters/combiners. The design and performance of these components are proprietary to LIGENTEC. The authors focused on designing two key components: the SiN phase shifter and the OPA antenna.

2.1. Phase shifter design

Phase shifters play a crucial role in OPA circuits by controlling the light phase at the input of each antenna, which allows the OPA to focus and steer the beam. The PS array allows the beam azimuthal steering, while beam steering in elevation is achieved through wavelength tuning of the input light.

The LIGENTEC platform used in this work includes a metal layer above the SiN, which can be utilized for thermo-optic phase shifting. SiN has a relatively low thermo-optic coefficient of $dn/dT = 2.45 \times 10^{-5} \text{ K}^{-1}$ [24], particularly when compared to materials like silicon, which has $dn/dT = 1.8 \times 10^{-4} \text{ K}^{-1}$ [25] at $\lambda = 1550 \text{ nm}$. Here, we present the design of a highly effective thermo-optic PS based on SiN.

The design process started by defining the dimensions of the metal tracks used as heating elements. The main limitation is the maximum current density in the metal film. Considering this restriction, the metal tracks are designed to have a width of $1.5 \mu\text{m}$ and a resistance of $71 \Omega/\text{mm}$. Simulations carried out in Ansys Lumerical Device [26] show that a single SiN waveguide covered with a single metal track (without recess) requires 320 mW of power to induce a 2π phase shift. To enhance PS efficiency and reduce power consumption, we employed two strategies:

1. Deep side etch: To minimize lateral heat dispersion, the oxide around the waveguide was etched down to the substrate, creating a recess (Fig. 2(a)). This recess is not filled with any other material, reducing thermal interference between neighboring PSs. The air filled recesses act as an effective heat insulator. The simulated heat distribution is shown in Fig. 2(b).
2. Double folded PS: This approach increases the concentration of heating elements within a tighter space. Instead of using a single long PS track, the track is folded on itself twice, resulting in the structure shown in Fig. 2(c).

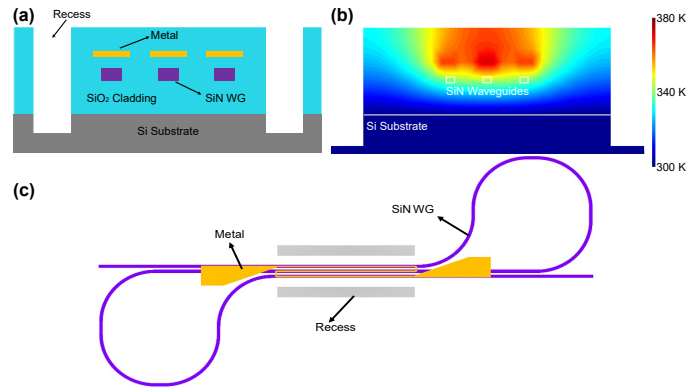


Fig. 2. (a) Double folded phase shifter cross-section. (b) Results of thermal simulations. (c) Top-view schematic of the phase shifter (not to scale).

According to simulations, we expect achieving an efficiency improvement of over 50% by implementing both these strategies. To verify the PS efficiency we used an imbalanced Mach-Zehnder interferometer-based test structure. The interferometer has two arms with different lengths to achieve a free spectral range (FSR) of approximately 5 nm, but with identical PS on both arms. Figure 3(a) shows the normalized transmission measured at the output of the interferometer in the case of a double folded PS. As expected, increasing the electrical power applied to the PS on one of the arms results in a red-shift in the wavelength of minimum transmission. A full FSR variation requires a 2π phase shift, which allows the extraction of the relationship between electrical power and phase shift. Figure 3(b) reports the power values required to induce a given phase shift for both the single-pass PS (dashed line) and the double-folded PS (solid line). The markers represent the measured values, while the lines are obtained by fitting. On average, the single pass PS with recess requires $245 \text{ mW}/2\pi$ of power, whereas the double folded PS requires only $130 \text{ mW}/2\pi$.

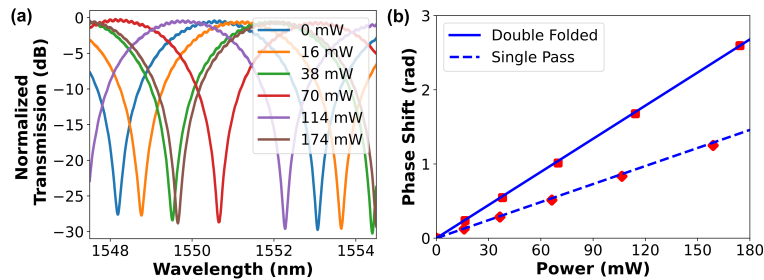


Fig. 3. (a) Optical transmission at different wavelengths and driving powers. (b) Required drive power for various phase shifts in radian.

The power consumption of the SiN OPA chip is mainly determined by the PS efficiency. The thermo-optic coefficient of Si is about one order of magnitude greater than that of SiN at 1550 nm, making a direct comparison of phase shifting efficiency between these materials imbalanced. However, several strategies employed in Si-based PSs can be adapted to further improve SiN PSs. One approach consists in increasing the waveguide density. As proved by [27], a closely packed Si waveguide spiral positioned beneath a single heating element can achieve improved efficiency. Additionally, the silicon substrate can be partially removed producing a suspended

structure which is thermally isolated from the rest of the chip [28,29], increasing the temperature in the waveguide area for the same electrical power input.

2.2. OPA antenna design

The antenna was designed using the approach proposed by Poulton in [13]. Figure 4(a) shows a top view schematic of the antennas. The perturbation is achieved by modulating the waveguide width through side etching. Figure 4(b) shows the relationship between perturbation dimension and strength. To ensure different parts of a single antenna emit at an identical elevation angle (ϕ), the perturbation period (Λ) is adapted accordingly. The antenna perturbation strength, and thus its dimension, is continuously varied along the length of the antenna to maintain a uniform power emission per unit length. In the proposed OPA the antenna is 1 mm long, and its perturbation dimension along its length is reported in Fig. 4(c). The antenna is designed to emit $0.95 \mu\text{W}/\mu\text{m}$ (95% of the input power) over its 1 mm length considering an input optical power of 1 mW. To evaluate the actual emission of the antenna, the chip contains dedicated test structures: four different antennas with lengths of $250 \mu\text{m}$, $500 \mu\text{m}$, $750 \mu\text{m}$ and 1 mm are connected to input and output grating couplers. The antenna emission can be considered as a loss if measured from the output grating coupler. Given the uniform power emission distribution along each antenna, the expected normalized losses would be $\sim 1.25 \text{ dB}$, $\sim 3 \text{ dB}$, $\sim 6 \text{ dB}$ and $\sim 13 \text{ dB}$, respectively. Figure 4(d) shows the measured normalized losses from the fabricated chip, which are lower than expected. By dividing the measured loss by the respective antenna's length and averaging the results, we demonstrate (Fig. 4(e)) the actual power emission is around $0.6 \mu\text{W}/\mu\text{m}$ with 1 mW input power. The lower than expected emission may be attributed to fabrication-related distortions of the perturbation, such as rounded edges rather than squared ones. However, this primarily affects the OPA's efficiency rather than its beam steering capabilities, which is the main objective of this work. The measured results are reported within the wavelength range of 1530 nm to 1570 nm , as grating coupler losses are relatively high outside this range, affecting the measurement's reliability.

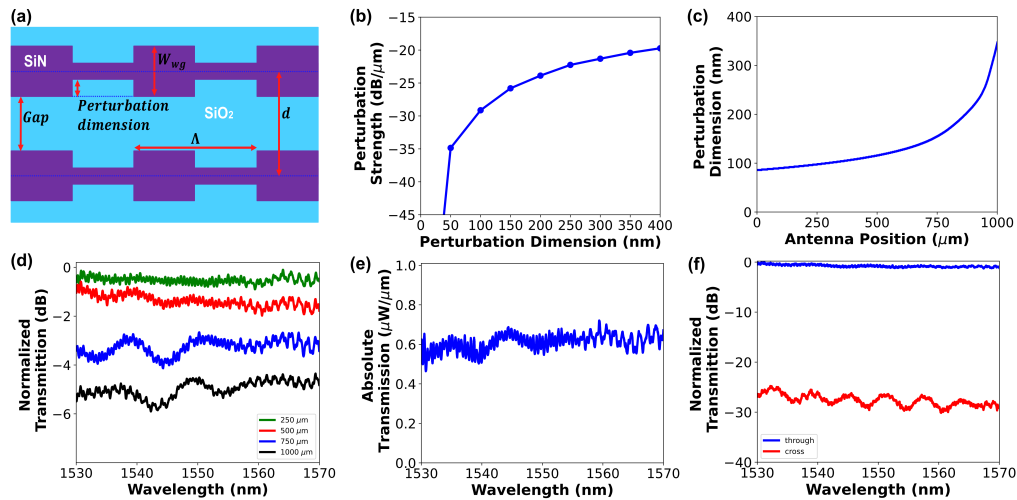


Fig. 4. (a) Top-view schematic of the OPA antennas. (b) Perturbation strength as a function of perturbation dimension. (c) Perturbation dimension along the antenna length. (d) Power transmission for antennas with different lengths at various wavelengths. (e) Absolute power transmission at various wavelengths. (f) Antennas cross-talk at varying wavelengths

The waveguide width (W_{wg}) and the gap between the antennas (Gap) were defined to ensure that power cross-talk between the antennas would be below 1%. Mode analysis [30] of the proposed

antenna array confirmed that a waveguide width of $W_{wg} = 1.2 \mu\text{m}$ and a gap of $Gap = 2.5 \mu\text{m}$ would meet this 1% cross-talk requirement for a 1 mm long antenna. Figure 4(f) shows the measured cross-talk between adjacent antennas to be lower than -26 dB.

With the antennas arranged in array of 32 elements, the OPA FOV and beam resolution (FWHM) can be theoretically calculated. The azimuth FOV is found using Eq. (1) [31]:

$$FOV = 2 \cdot \arcsin\left(\frac{\lambda}{2d}\right). \quad (1)$$

and for this work it is equal to 24.2° . In this equation λ is the wavelength, and d is the pitch of the OPA channels. Elevation steering (ϕ) of the beam is possible by tuning the wavelength of the input laser and is expressed by the Bragg condition concept [32]:

$$\sin \phi = \frac{\Lambda n_{eff} - \lambda}{\Lambda n_c}. \quad (2)$$

in which ϕ is the elevation steering angle, Λ is the period of the antenna, n_{eff} is the effective refractive index of the emitting antenna, and n_c is the top cladding (SiO_2) refractive index. For a wavelength range of 1500 nm to 1630 nm, an elevation FOV of 9.57° is determined.

As for the beam resolution along azimuth and elevation directions, the FWHM of the beam is given by:

$$FWHM = \frac{2.78 \lambda}{\pi A}. \quad (3)$$

which indicates that the beam size has dependency on the OPA aperture size (A) along each direction and wavelength [13]. We have obtained a resolution of 0.66° and 0.08° for azimuth and elevation directions, respectively.

2.3. OPA chip assembly

Four different OPAs were designed and integrated onto a single $10 \text{ mm} \times 5 \text{ mm}$ chip. These designs vary in terms of the number of antennas, waveguide width, length, and gap. As mentioned earlier, the findings presented in this work pertain specifically to the 1×32 OPA with an antenna width of $W_{wg} = 1.2 \mu\text{m}$, and $Gap = 2.5 \mu\text{m}$. Each OPA on the chip features a dedicated input edge coupler, positioned at the bottom of the circuit. Each PS is electrically connected to a dedicated set of pads located at the edge of the chip.

To perform the final testing, the chip was first glued to a custom-designed aluminum plate. A thermistor was placed beneath the chip to provide feedback for temperature control via a thermoelectric cooler (TEC) positioned under the aluminum plate. A printed circuit board (PCB) was designed, manufactured, and mounted around the chip to facilitate the delivery of input control signals for the PS array from ribbon cables. Each pad on the chip was connected to a corresponding pad on the PCB, with wire diameters chosen to meet maximum current requirements.

As previously described, our chip incorporates four distinct OPAs along with various test structures. These include structures designed for measuring PBS and edge coupler losses, as well as for the alignment of fiber array itself. In total, the chip features 16 input channels (edge couplers) to couple light to each individual structure. The 16-channel fiber array was actively aligned and fixed to the chip, using an ultraviolet-cured epoxy. The bonding of the chip to the aluminum holder and the fiber attachment were performed by CamGraPhIC Srl in Pisa, Italy. Figure 5(c) shows the chip in its final assembly.

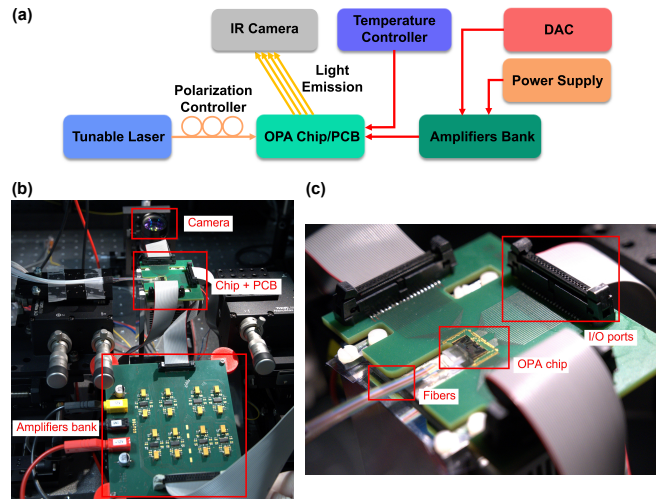


Fig. 5. (a) Schematic of the beam steering setup. (b) Experimental setup for beam characterization. (c) Closer look of the OPA chip and the PCB.

3. OPA characterization

3.1. Characterization setup

The setup used to characterize the OPA's beam forming and steering capabilities is illustrated in Fig. 5. It consists of a tunable laser, fiber, polarization controller, OPA chip, PCB, power supply, digital-to-analog converter (DAC), bank of amplifiers, temperature controller, screen, InGaAs IR camera, and a computer.

At a given wavelength, the laser injects light into the fiber. The light passes through a polarization controller to maximize the power in the TE component. The TE polarized light then enters the OPA chip through the edge coupler and is distributed among 32 different channels. Before reaching the antenna, the signals' phases are adjusted by the PSs. A computer controls a programmable analog signal generator, i.e. DAC, with dedicated outputs to each PS. Since the DAC output current is limited, a bank of current amplifiers is employed to support it. The PCB, to which the OPA chip is wire-bonded, is connected to the bank of amplifiers via ribbon cables. Figure 5(b) and (c) provide more detailed illustrations of the setup.

Once the light signals in all antennas are in phase, the light emitted by the array of antennas interfere constructively in a specific direction. The camera in the far-field measurement setup captures the diffracted light from the reflective screen, mounted far from the OPA chip. The acquired light intensity is then saved as a function of the emission angle.

3.2. Beam optimization

In the uniform array of optical antennas (equidistant antennas), the beam is azimuthally steerable by applying an $(i - 1)\alpha$ phase shift to each antenna, where i and α are the number of the antennas and the constant phase shift among neighboring antennas, respectively. This is shown by [31]

$$\sin \theta = \frac{\lambda \alpha}{2\pi d}. \quad (4)$$

where θ is the beam's azimuth steering angle, λ is the wavelength, and d is the pitch of the OPA channels. In practice, due to unwanted and accumulated phase noise in the path from the splitters to the antennas, the beam cannot be steered and shaped by applying a constant phase difference α between them. Hence, it is essential to implement an efficient algorithm to choose the optimum

phase shift for each individual PS and to correct the phase variation. Hill-climbing, particle swarm optimization, and gradient descent algorithms are the common techniques to calibrate the OPA functionality [33,34]. Additionally, phase interrogators and on-chip photodetectors have been employed to correct the consequent phase accumulations in some studies [33,34].

The beam steering algorithm used in this work is based on available Python libraries integrated with a custom code developed by the authors. The algorithm selects the optimum driving signal (voltage) for the PSs array to achieve a single spot on the screen. This selection process is repeated for each wavelength and for each point in the azimuth FOV. Figure 6 illustrates how the beam steering algorithm operates, running on a PC to ensure the synchronous operation of all devices.

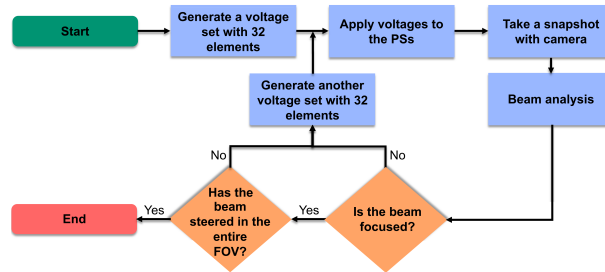


Fig. 6. Beam steering algorithm flow chart.

The output combiner reported in Fig. 1(a) has been introduced in the circuit as a potential fallback strategy for beam calibration. In our design, all 32 channels have equal lengths before reaching the combiner. By singularly driving each PS in the array and measuring the power from the output coupler array, it would have been possible to characterize each PS and use this data to drive the OPA. Given the successful calibration using the screen approach, this second strategy was not investigated further.

3.3. Results and discussion

In Fig. 7(a), a selection of the formed beams during the azimuthal beam steering at different wavelengths are illustrated. The bottom line corresponds to an azimuthal beam steering at a wavelength of 1630 nm, the middle to 1550 nm, and the top to 1500 nm. At each wavelength, the beams located at the far left and far right represent the grating lobes, occurring simultaneously. The angular distance between these two points defines the azimuthal FOV at each specific wavelength, while the beams within this range correspond to the steered beams. The azimuth FOV at 1550 nm is measured to be 24.28° , which is consistent with the expected value shown earlier. As illustrated in Fig. 7(a), the FOV increases with increasing wavelength, as predicted by Eq. (1). Furthermore, the measured elevation FOV derived by wavelength tuning is 9.46° , which aligns with the theoretical data.

Additionally, the spot size of the emitted beam was characterized. As shown in Fig. 7(b), a resolution of 0.66° and 0.082° have been measured for the beam at 1550 nm along azimuth and elevation directions, respectively. These results confirm that the OPA delivers the specifications it was designed for. Table 1 compares the results of this work with other OPAs integrated on different platforms. The beam intensity measurement in the azimuth direction is demonstrated in Fig. 7(c). A side lobe suppression of 8.62 dB has been achieved through this study.

SiN lower refractive index in respect to silicon highly impacts the minimum pitch between the antennas, and consequently the OPA azimuth FOV. Other approaches can be implemented to increase the OPA FOV such as: non-uniform OPAs [27] or a trapezoidal slab grating [11]. In the first approach, Si antenna elements are located farther apart (well beyond the cross-talk distance

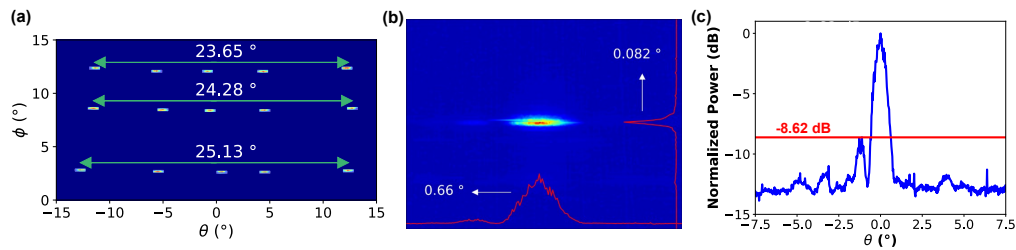


Fig. 7. (a) 2D beam steering from 1500 nm to 1630 nm (see Visualization 1). (b) Beam resolution at 1550 nm along azimuth (0.66°) and elevation (0.082°) directions. (c) Azimuthal far-field intensity at 1550 nm with side lobe suppression of -8.62 dB.

Table 1. Comparison between OPA performances based on different platforms

Platform	Channels	λ	FOV	Spot size	PS efficiency
This work	32	1500-1630 nm	$24.28^\circ \times 9.46^\circ$	$0.66^\circ \times 0.082^\circ$	65 mW/ π
SiN [22]	32	905 nm (fixed)	$34^\circ \times 12^\circ$	$1.3^\circ \times 0.2^\circ$	30 mW/ π
SiN [8]	16	905 nm (fixed)	$17.6^\circ \times 3^\circ$	$4.3^\circ \times 0.7^\circ$	87.6 mW/ π
SiN-on-SOI [29]	128	1510-1630 nm	$160^\circ \times 3^\circ$	$0.16^\circ \times 0.04^\circ$	6 mW/ π
SiN-on-SOI [35]	256	1470-1570 nm	$140^\circ \times 16.7^\circ$	$0.051^\circ \times 0.016^\circ$	NA
SiN-on-SOI [9]	64	1500-1630 nm	$35.5^\circ \times 22.7^\circ$	$0.69^\circ \times 0.075^\circ$	17.5 mW/ π
SOI [27]	120	1480-1572 nm	$162^\circ \times 13.7^\circ$	$0.13^\circ \times 0.13^\circ$	3.1 mW/ π
SOI [36]	512	1450-1640 nm	$56^\circ \times 15^\circ$	$0.04^\circ \times 0.04^\circ$	1 μ W/ π

$\sim 50 \mu\text{m}$) achieving an FOV of 162° . In the latter approach, an array of Si waveguides with varying widths is connected to a trapezoidal vertically emitting slab grating resulting in an FOV of 180° . Similar strategies may be applied to an OPA fully integrated on SiN to improve its FOV.

4. Conclusions

In this paper, we reported the design, fabrication, and testing of a 1×32 channel SiN OPA. The design was based on a commercially available platform, incorporating antennas and dedicated highly efficient thermo-optic phase shifters. Additional components such as edge couplers, polarization beam splitters, and splitters were taken from the available PDK. The proposed phase shifter exhibited a measured efficiency of 65 mW/ π . Characterization of the assembled OPA proved an FOV of $24.28^\circ \times 9.46^\circ$ and a beam FWHM of $0.66^\circ \times 0.082^\circ$, surpassing the current state-of-the-art in fully SiN-based OPAs. This work showcases the potential of SiN-based OPAs for a wide range of applications that require beam forming and steering capabilities across a broad spectral range.

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Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

References

1. R. Osgood jr and X. Meng, "Integrated optics: An overview," *Principles of Photonic Integrated Circuits: Materials, Device Physics, Guided Wave Design* **1**, 1–9 (2021).
2. Z. Yan, Y. Han, L. Lin, *et al.*, "A monolithic InP/SOI platform for integrated photonics," *Light: Sci. Appl.* **10**(1), 200 (2021).
3. C. Xiang, W. Jin, and J. E. Bowers, "Silicon nitride passive and active photonic integrated circuits: trends and prospects," *Photonics Res.* **10**(6), A82–A96 (2022).

4. Y. Jiang, X. Han, H. Huang, *et al.*, “Monolithic photonic integrated circuit based on silicon nitride and lithium niobate on insulator hybrid platform,” *Adv. Photonics Res.* **3**(10), 2200121 (2022).
5. C.-S. Im, B. Bhandari, S.-M. Kim, *et al.*, “Backward-emitting silicon nitride optical phased array enabling efficient wavelength-tuned beam steering,” *IEEE Photonics J.* **12**(5), 1–10 (2020).
6. P. Wang, G. Luo, Y. Xu, *et al.*, “Design and fabrication of a SiN-Si dual-layer optical phased array chip,” *Photonics Res.* **8**(6), 912–919 (2020).
7. C.-S. Im, B. Bhandari, K.-P. Lee, *et al.*, “Silicon nitride optical phased array based on a grating antenna enabling wavelength-tuned beam steering,” *Opt. Express* **28**(3), 3270–3279 (2020).
8. N. A. Tyler, D. Fowler, S. Malhouitre, *et al.*, “SiN integrated optical phased arrays for two-dimensional beam steering at a single near-infrared wavelength,” *Opt. Express* **27**(4), 5851–5858 (2019).
9. Q. Wang, S. Wang, L. Jia, *et al.*, “Silicon nitride assisted 1 × 64 optical phased array based on a SOI platform,” *Opt. Express* **29**(7), 10509–10517 (2021).
10. T. C. Cheston and J. Frank, “Phased array radar antennas,” *Radar Handbook* **2**, 1 (1990).
11. Y. Liu and H. Hu, “Silicon optical phased array with a 180-degree field of view for 2D optical beam steering,” *Optica* **9**(8), 903–907 (2022).
12. Y. Li, B. Chen, Q. Na, *et al.*, “Wide-steering-angle high-resolution optical phased array,” *Photonics Res.* **9**(12), 2511–2518 (2021).
13. C. V. Poulton, “Integrated LIDAR with optical phased arrays in silicon photonics,” Ph.D. thesis, Massachusetts Institute of Technology (2016).
14. I. Kim, R. J. Martins, J. Jang, *et al.*, “Nanophotonics for light detection and ranging technology,” *Nat. Nanotechnol.* **16**(5), 508–524 (2021).
15. T. Sharma, J. Wang, B. K. Kaushik, *et al.*, “Review of recent progress on silicon nitride-based photonic integrated circuits,” *IEEE Access* **8**, 195436–195446 (2020).
16. N. Li, C. P. Ho, J. Xue, *et al.*, “A progress review on solid-state LiDAR and nanophotonics-based LiDAR sensors,” *Laser & Photonics Reviews* **16**(11), 2100511 (2022).
17. D. Bastos, P. P. Monteiro, A. S. Oliveira, *et al.*, “An overview of LiDAR requirements and techniques for autonomous driving,” in *2021 Telecoms Conference (ConfTELE)*, (IEEE, 2021), pp. 1–6.
18. C. Xiang, J. Guo, W. Jin, *et al.*, “High-performance lasers for fully integrated silicon nitride photonics,” *Nat. Commun.* **12**(1), 6650 (2021).
19. H. Nejadriahi, S. Pappert, Y. Fainman, *et al.*, “Efficient and compact thermo-optic phase shifter in silicon-rich silicon nitride,” *Opt. Lett.* **46**(18), 4646–4649 (2021).
20. H. Qiu, Y. Liu, X. Meng, *et al.*, “Bidirectional high sidelobe suppression silicon optical phased array,” *Photonics Res.* **11**(4), 659–668 (2023).
21. H. Xu and Y. Shi, “Diffraction engineering for silicon waveguide grating antenna by harnessing bound state in the continuum,” *Nanophotonics* **9**(6), 1439–1446 (2020).
22. D. Fowler, S. Guerber, J. Faugier-Tovar, *et al.*, “SiN-based integrated optical phased array for 2D beam steering at 905 nm,” in *2023 IEEE Photonics Conference (IPC)*, (IEEE, 2023), pp. 1–2.
23. “LIGENTEC | Photonic Integrated Circuits Manufacturing - LIGENTEC — ligentec.com,” <https://www.ligentec.com/>. [Accessed 19-04-2024].
24. A. Arbabi and L. L. Goddard, “Measurements of the refractive indices and thermo-optic coefficients of Si₃N₄ and SiO_x using microring resonances,” *Opt. Lett.* **38**(19), 3878–3881 (2013).
25. J. Komma, C. Schwarz, G. Hofmann, *et al.*, “Thermo-optic coefficient of silicon at 1550 nm and cryogenic temperatures,” *Appl. Phys. Lett.* **101**(4), 041905 (2012).
26. “Lumerical HEAT — Intro | Ansys Innovation Courses — innovationspace.ansys.com,” <https://www.ansys.com/products/optics/mode#tab1-1>. [Accessed 30-04-2024].
27. H. Qiu, Y. Liu, X. Meng, *et al.*, “Ultra-sparse aperiodic silicon optical phased array using high-performance thermo-optic phase shifter,” *Laser & Photonics Reviews* **18**(10), 2301177 (2024).
28. Q. Fang, J. F. Song, T.-Y. Liow, *et al.*, “Ultralow power silicon photonics thermo-optic switch with suspended phase arms,” *IEEE Photonics Technol. Lett.* **23**(8), 525–527 (2011).
29. B. Chen, Y. Li, Q. Xie, *et al.*, “SiN-On-SOI optical phased array lidar for ultra-wide field of view and 4D sensing,” *Laser & Photonics Reviews* **18**(10), 2301360 (2024).
30. “Ansys Lumerical MODE Optical Waveguide and Coupler Design Software — ansys.com,” <https://www.ansys.com/products/optics/mode#tab1-1>. [Accessed 30-04-2024].
31. J. He, T. Dong, and Y. Xu, “Review of photonic integrated optical phased arrays for space optical communication,” *IEEE Access* **8**, 188284–188298 (2020).
32. Y. Xu, F. Wang, Y. Gao, *et al.*, “Efficient polymer waveguide grating coupler with directionality enhancement,” *Opt. Commun.* **463**, 125418 (2020).
33. J. Ø. Kjellman, M. Prost, A. Marinins, *et al.*, “Densely integrated phase interrogators for low-complexity on-chip calibration of optical phased arrays,” *J. Lightwave Technol.* **40**(16), 5660–5667 (2022).
34. M. Prost, J. Ø. Kjellman, S. Dwivedi, *et al.*, “Optical phased array with on-chip phase calibration,” *Opt. Lett.* **47**(23), 6081–6084 (2022).
35. W. Xu, Y. Guo, X. Li, *et al.*, “Fully integrated solid-state lidar transmitter on a multi-layer silicon-nitride-on-silicon photonic platform,” *J. Lightwave Technol.* **41**(3), 832–840 (2023).
36. C. V. Poulton, M. J. Byrd, P. Russo, *et al.*, “Long-range lidar and free-space data communication with high-performance optical phased arrays,” *IEEE J. Sel. Top. Quantum Electron.* **25**(5), 1–8 (2019).